Yoshi

Bench upholstery & cushion construction plan. Platform cushion method with channel pleating, moisture barriers, and fold-to-bed conversion.

18″×48″ FULL BENCH

16″×48″ BED DEPTH

40″×48″ MATTRESS (W×FRAME)

3″ FOAM DEPTH

01 Upholstery Vocabulary

STYLEBOOK

CHANNELS
Parallel ridges sewn through fabric + sew foam at regular intervals. Also called pleating or tuck & roll.

SEW FOAM / SCRIM
Thin foam (1/2″–1/4″) with fabric backing. Sewn behind your fabric to create channel puffiness.

BOXING
Strip of fabric wrapping around a cushion's sides — the "walls" of the cushion.

PLATES
The flat top and bottom panels of a cushion cover.

BATTING / DACRON
Polyester wrap around foam. Adds plumpness, softens edges, protects from UV.

PLATFORM CUSHION
Fabric stapled to backer board, no bottom plate. Ideal for secured bench seating.

BELLBLOC 68
Non-woven hydrophobic PET cushion liner. Moisture, mold, mildew barrier. Adds body to soft fabrics.

SILK FILM
Thin nylaseles plastic film wrapped around foam. Waterproof barrier + eases foam insertion.

PIPING / WELTING
Cord wrapped in fabric, sewn into seams. Adds a finished, structural edge.

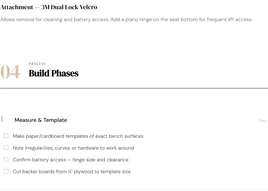
CAMBRIC
Dust cover fabric stapled to bottom, hiding raw edges and staples.

02 Bench Dimensions

SPECIFICATIONS

COMPONENT	DEPTH	WIDTH	NOTES
Bench Bottom	18″	48″	Covers house batteries · sits on bench frame
Bench Back	16″	48″	Folds for bed conversion
Mattress Panels	40″	48″	Future phase · connects when flat

FIG. A — CUSHION CROSS-SECTION



03 Construction Approach

DESIGN DECISIONS

SELECTED
Platform Cushion — Stapled to Backer Board

No zippers. Fabric stapled directly to ¾″ plywood backer. Can be hinged or Velcro'd for battery access. Foam glued to board, wrapped in batting, wrapped in channel fabric, stapled to back.

SELECTED
Channel Pleating — 9 Channels at 5″ Spacing

Channels prevent fabric shifting, add structural memory, distribute wear evenly, and give your soft fabric the body it needs. Sew straight lines through fabric + BellBloc + sew foam. Your Singer HD handles this.

FIG. B — CHANNEL LAYOUT (TOP VIEW)

SELECTED
Foam — 3″ Seat , 2″ Back, High Density

1.8+ lb density for seat bottom. Medium density for back. Future mattress panels 3″–4″ or latex for sleep comfort.

> **MOISTURE STRATEGY — DUAL BARRIER**
>
> In a van, condensation, spills, and rain from open doors are constant threats. BellBloc 68 between fabric and batting handles moisture, mold, and mildew while adding body to your soft fabric — likely eliminating the need for separate interfacing. Silk film directly on foam is the nuclear option. Between both layers plus topical Scotchgard, your foam is extremely well protected.

SELECTED
Attachment — 3M Dual Lock Velcro

Allows removal for cleaning and battery access. Add a piano hinge on the seat bottom for frequent lift access.

04 Build Phases

BUILDING

1 Measure & Template — Day 1

- [ ] Make paper/cardboard templates of exact bench surfaces
- [ ] Note irregularities, curves, or hardware to work around
- [ ] Confirm battery access — hinge side and clearance
- [ ] Cut backer boards from ¾″ plywood to template size

2 Foam Prep — Day 2

- [ ] Cut foam ¼″ larger than backer board on all sides
- [ ] Spray-glue foam to backer board (3M Super 77)
- [ ] Wrap foam in Cushion Wrap Silk Film
- [ ] Wrap batting over silk film, pull snug, staple to back of board

3 Channel Sewing — Day 3

- [ ] Lay fabric right-side-down → BellBloc → sew foam (scrim up)
- [ ] Mark channel lines with chalk at 5″ intervals
- [ ] Sew straight lines through all three layers
- [ ] Account for ~1″–2″ width shrinkage from puffing

4 Wrap & Staple — Day 3-6

- [ ] Lay channeled fabric face-down, place assembly on top
- [ ] Pull fabric to back, starting from center of each side
- [ ] Staple every 1″–2″ — stop 4″ from corners
- [ ] Wrap corners: center fold, then two side folds
- [ ] Trim excess, staple cambric dust cover to back

5 Protect & Install — Day 6

- [ ] Apply Scotchgard — 2 coats, dry between
- [ ] Attach Dual Lock Velcro to backer board and bench
- [ ] Install hinges if needed for battery access
- [ ] Test fit, adjust, admire

05 Cut Dimensions

CUTTING

Seat Bottom

```
BACKER BOARD 17.5″ × 47.5″ FOAM 18″ × 48″ × 3″ 5″ larger than board each side BATTING 20″ × 50″ wraps
fully + 2″ overlap FABRIC (flat) 30″ × 60″ 48 × (3×2) + (3×2) staple flap FABRIC (cut) 30″ × 60″ ×9″
take-up (9 channels × ~2″) BELLBLOC same as fabric SCR FOAM 18″ × 48″ cushion face only
```

Seat Back

```
BACKER BOARD 15.5″ × 47.5″ FOAM 16″ × 48″ × 2″ BATTING 24″ × 56″ FABRIC (flat) 24″ × 56″×60″ FABRIC
(cut) add 3″×9″ to width BELLBLOC same as fabric SCR FOAM 16″ × 48″ CAMBRIC backer + 1″ fold each side
```

06 Shopping List

BUYING

FOAM & PADDING		
High-density foam 3″ 1.8+ lb density	18″ × 48″	$25–40
Medium-density foam 2″ Back cushion	16″ × 48″	$15–25
Sew foam / scrim ¼″ Fabric-backed for channels	1 yard	$8–15
Polyester batting Thicker bonded type	2–3 yards	$10–15/yd
BellBloc 68 Moisture barrier liner	2–3 yards	$8–12/yd
Cushion Wrap Silk Film Waterproof foam wrap	2 yards	$8–12
ADHESIVES & PROTECTION		
3M Super 77 Foam-safe spray adhesive	1 can	$12–15
Scotchgard / 303 Fabric Guard	1 can	$10–15
HARDWARE & NOTIONS		
¾″ plywood backer boards	2 pcs	$10–20
½″ staples (T50)	1 box	$5–8
3M Dual Lock / Velcro	2–3 ft	$10–15
Piano hinge If needed for battery access	48″	$8–12
Cambric dust cover	1 yard	$5–8
THREAD & NEEDLES		
Polyester thread Tex 70 Match to fabric color	1 spool	$8–12
HD needles #16 or #18	1 pack	$5–8

> **ESTIMATED TOTAL**
>
> $150–250 depending on foam quality and optional items. Your fabric and Singer HD are already significant assets.

Sailrite sailrite.com Best overall — foam, thread, notions, tutorials	**Action Upholstery** actionupholstery.com BellBloc 68 by the yard	**Outdoor Textiles** outdoortextiles.com BellBloc, marine upholstery
Albright's Supply albrightsupply.com Sew foam, auto upholstery tools	**Amazon** amazon.com Foam/touch, batting, basics	**Home Depot / Lowe's** Plywood, staple guns, hinges

07 Tutorial Watch List

EDUCATION

1. **Sailrite — Platform Cushion** CORE
   Your exact project. Covers measuring, cutting, wrapping, stapling, piping.
   `sailrite.com/platform-cushion`

2. **Sailrite — Pleated Motorcycle Seat** CHANNELS
   How to sew channels using sew foam/scrim. Same technique regardless of context.
   `sailrite.com/…Pleated-Seat-Cover-Motorcycle`

3. **Sailrite — Reupholster a Truck Seat**
   Full truck bench with channeled inserts. Most directly relevant to your build.
   `sailrite.com/…Reupholster-Truck-Seat`

4. **Sailrite — Vinyl Box Cushion with Channeling**
   Channeling sewn into cushion panels + piping. Good for leveling up.
   `sailrite.com/how-to-make-box-cushion`

5. **Sailrite — Folding Cockpit Cushions** BED PHASE
   Hinged cushion that folds. Your guide for the future mattress conversion.
   `sailrite.com/diy-folding-cockpit-cushion`

6. **Sailrite — Protecting Foam from Moisture & UV**
   Van-critical info on silk film, batting, and moisture barriers.
   `sailrite.com/protect-foam-water-sun-light-damage`

7. **Instructables — Cushions for a Camper Van**
   Step-by-step with photos from a real van conversion.
   `instructables.com/…sew-cushions-camper-van`

8. **Campervan Converts — Seat Cushions for Beginners**
   Beginner-friendly with good corner techniques.
   `campervanconverts.com/…beginner-part-2`

YOUR FABRIC

08 Tips for Soft, Drapey Textiles

01 **It will shift while sewing.** Use lots of pins or basting tape. Sew slowly.

02 **BellBloc 68 likely replaces iron-on interfacing** — adds body and moisture protection in one layer. Try without separate interfacing first.

03 **Double the batting layer on the seat** to compensate for fabric thinness.

04 **Channels are your friend.** They give this fabric the structure it doesn't naturally have.

05 **Pre-shrink your fabric** if washable. Better now than after it's stapled.

06 **Scotchgard liberally.** BellBloc protects the foam; spray protects the fabric face.

07 **Test Singer tension on scrap** — fabric + BellBloc + sew foam before committing.

SUMMARY

09 Build Order

1. **Watch** Sailrite Platform Cushion + Motorcycle Pleating tutorials
2. **Order** foam, batting, BellBloc 68, sew foam, silk film, thread, staples, Velcro
3. **Template** the bench while supplies ship
4. **Cut** backer boards and foam
5. **Practice** channel sewing on scrap (test strip!)
6. **Build seat bottom first** — the one you'll use most
7. **Build seat back** — apply lessons learned
8. **Install, test, adjust**
9. **Plan mattress panels** — phase 2 bed conversion

> FUTURE — BED CONNECTION
>
> Seat bottom and mattress panels at the same height when flat. See Sailrite folding cockpit cushion tutorial for hinge technique. Velcro strips along mating edges for alignment. Same channel pattern for visual continuity. Mattress foam: 3"–4" medium-firm or 2" latex topper.